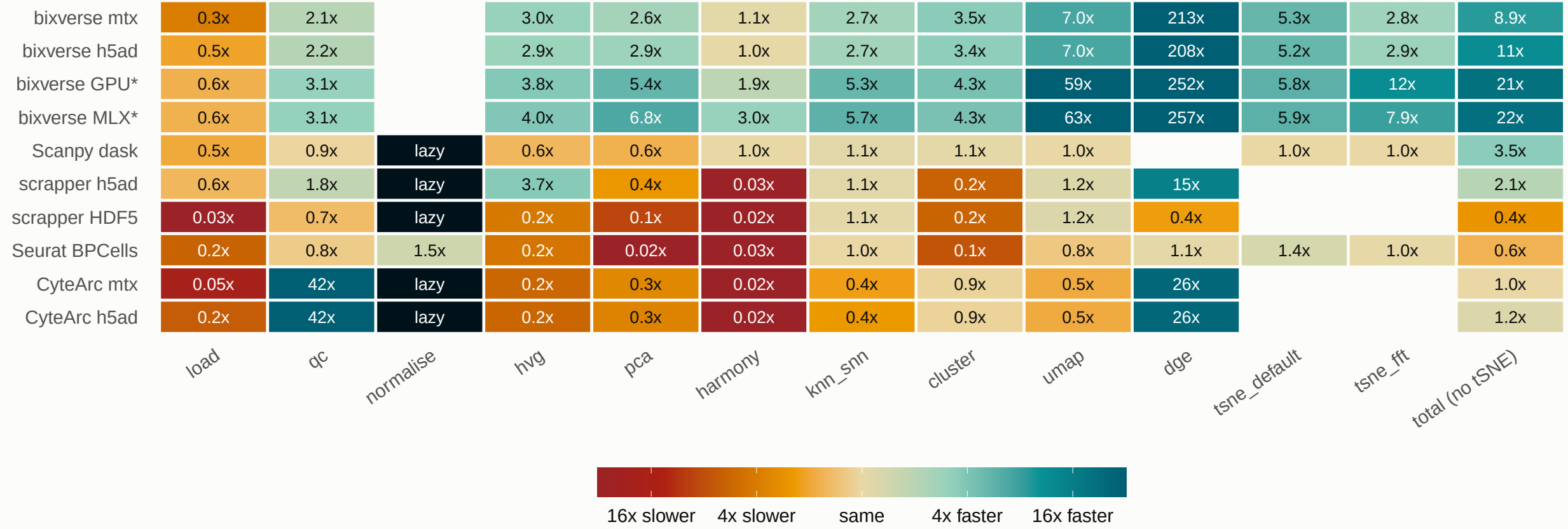
## Speed−up per step against in−memory Scanpy (h5ad), 500k

The largest size in−memory Scanpy survives. Only runs that completed at 500k.




bixverse normalises during load. Scanpy dask has no DGE step. bixverse CPU tSNE builds its kNN with IVF.
tsne_quick (qdtsne) and cluster_fast (Leiden on k−means centroids) are not shown and not in the total.